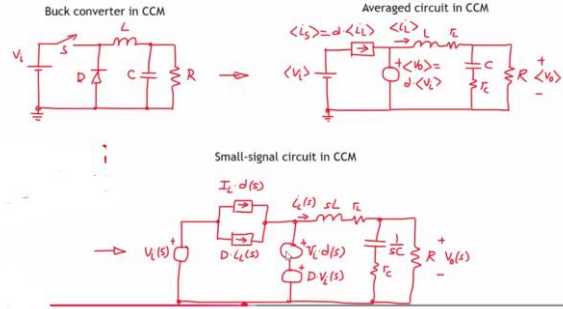


Input Impedance

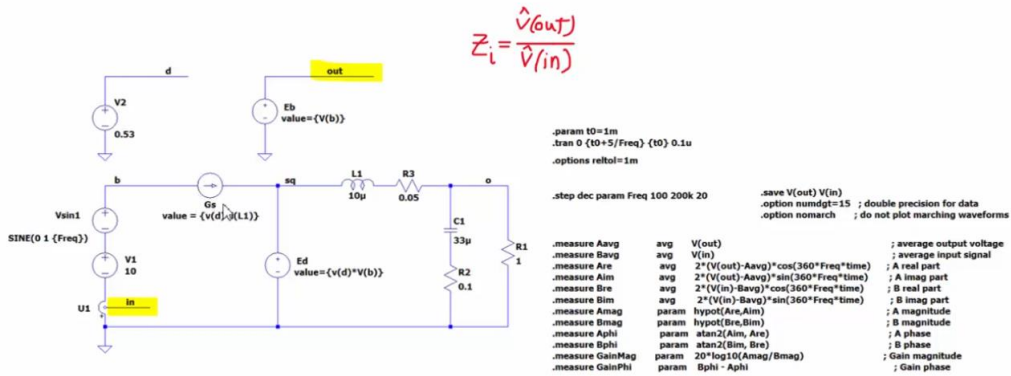
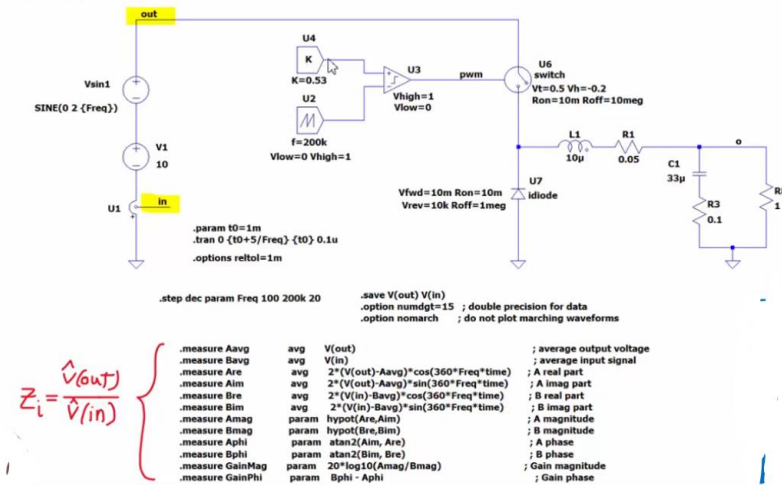
LT#21

PE#41

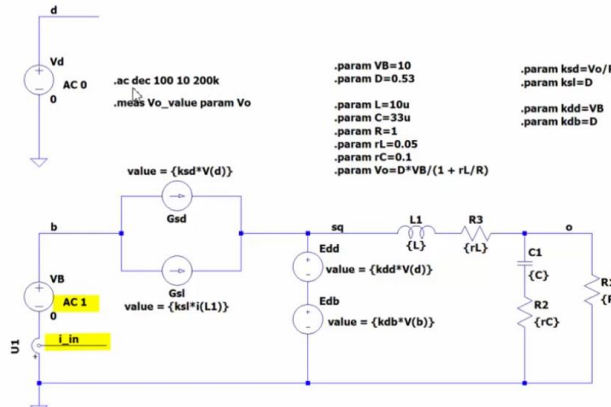
Open Loop : Input Impedance



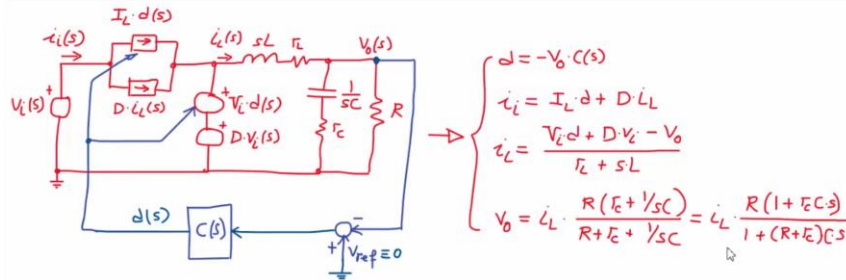
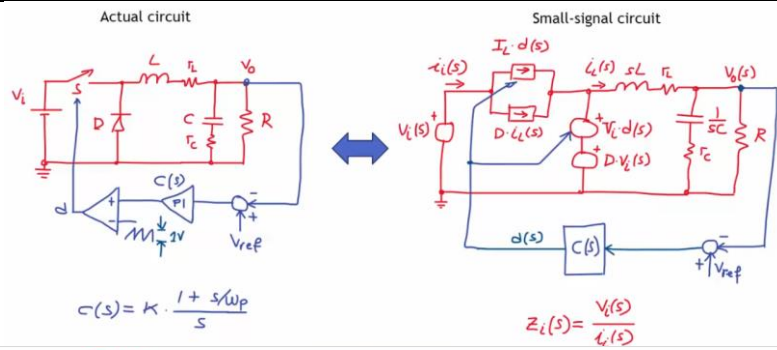
First way: Using the actual circuit



$$Z_i = \frac{1V}{v(i-in)}$$



Closed Loop : Input Impedance



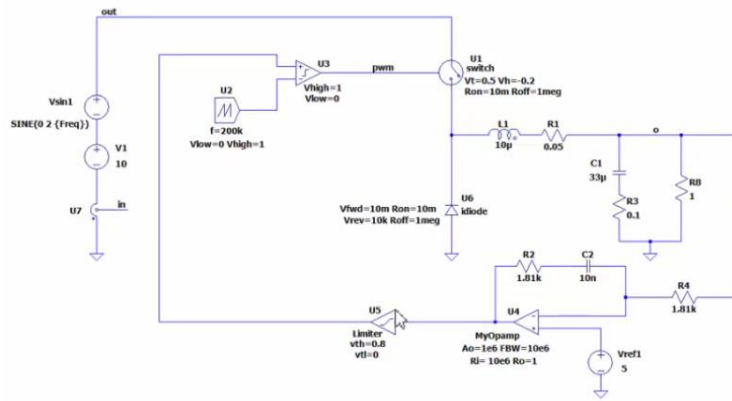
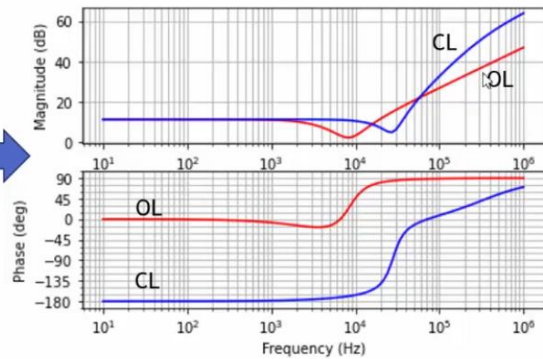
$$Z_i(s) = \frac{V_i(s)}{I_i(s)} = \frac{1}{D} \cdot \frac{[1 + (R + r_L)Cs](r_L + sL) + [V_L \cdot C(s) + 1]R \cdot (1 + r_L C s)}{D[1 + (R + r_L)Cs] - I_L \cdot C(s)R(1 + r_L C s)}$$

Assuming: $C(0) = \infty \rightarrow Z_i(0) = \frac{-V_L}{D \cdot I_L} = \frac{-V_L}{D \cdot I_o} \quad ; \quad Z_i(\infty) = \infty$

```

1 # -*- coding: utf-8 -*-
2 from math import pi, log10, sqrt
3 from control import tf, bode_plot
4
5 L=10e-6; rL=0.05; C=33e-6; rC=0.1; R=1;
6 Vi=10; D=0.53;
7 # Output voltage and current
8 Vo=D*Vi/(1 + rL/R)
9 IL=Vo/R
10 # PI compensator
11 R1=1.81e3; R2=1.81e3; C2=10e-9
12 wz=1/(R2*C2)
13 s = tf('s')
14 Zi_ol= (R/(D*R*C)*s*(L*(1+rC/R)*s**2 + (L/R+rC*C+rL*L*rC/R)*s + 1+rL/R) / \
15 (1 + (R+rC)*C*s)
16 Zi_cl= (1/D)*(1+(R+rC)*C*s)*(rL+L*s) + (Vi*(1+s/wz)/(s/wz)+1)*R*(1+rC*C*s) /\
17 (D*(1 + (R+rC)*C*s) - IL*(1+s/wz)/(s/wz)*R*(1+rC*C*s))
18 # Plot Plant's Bode
19 # Note that once Hz is true, omega limits are in Hz
20 mag, phase, omega = bode_plot(Zi_ol, dB=True, Hz=True, omega_limits=(10,1000e3), \
21 omega_num=100, color="red")
22 mag, phase, omega = bode_plot(Zi_cl, dB=True, Hz=True, omega_limits=(10,1000e3), \
23 omega_num=100, color="blue")
24

```



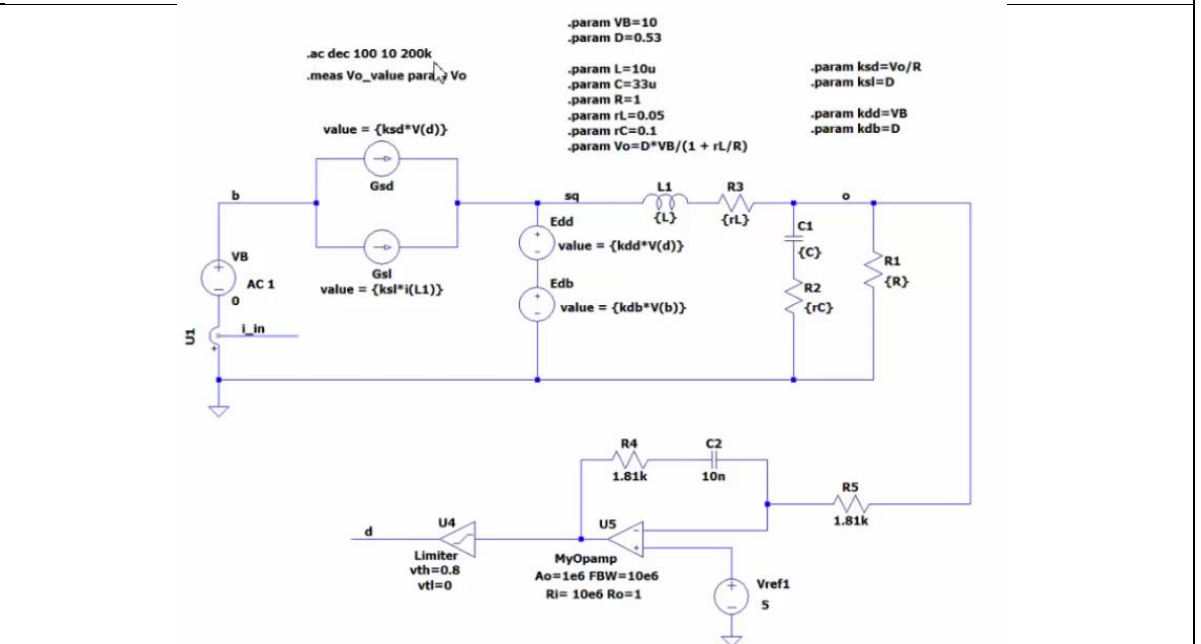
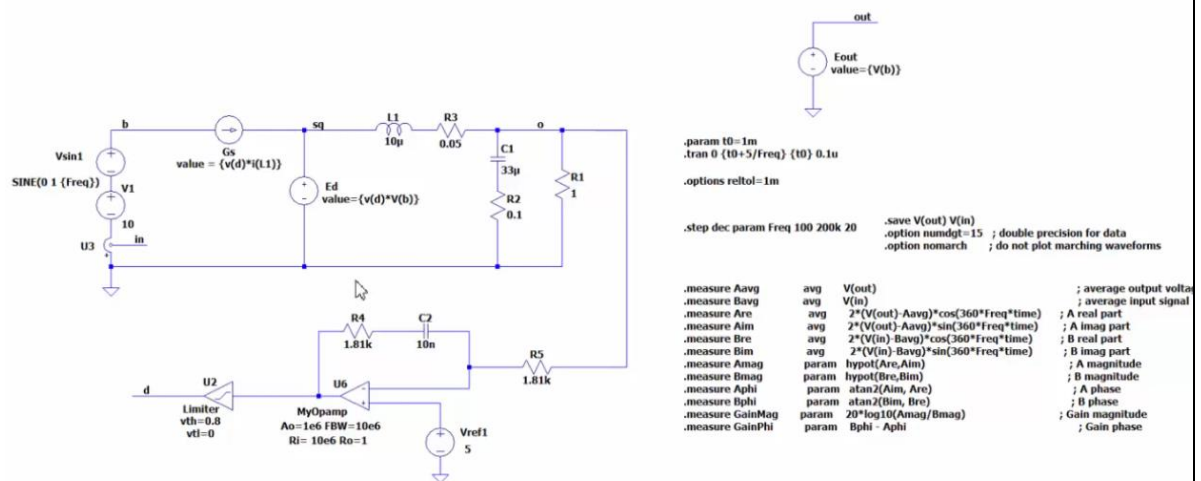
```

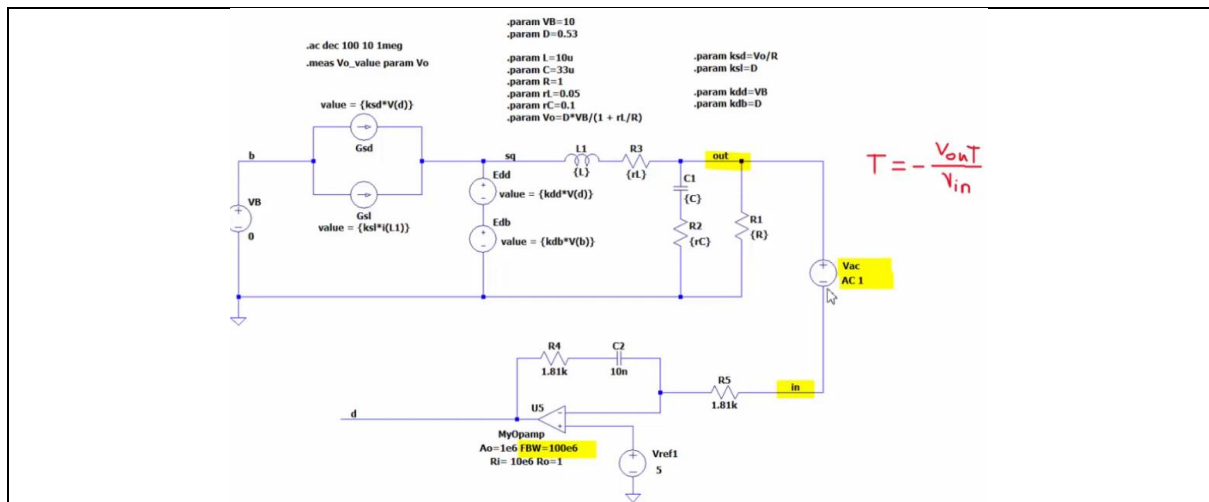
.param t0=1m
.tran 0 (t0+5/Freq) (t0) 0.02u uk
.options retol=1

.step dec param Freq 100 200k 25
.save V(out) V(in)
.option numdgt=15 ; double precision for data
.option nomarch ; do not plot marching waveforms

.measure Aavg avg V(out) ; average output voltage
.measure Bavg avg V(in) ; average input signal
.measure Areal 2*(V(out)-Aavg)*cos(360*Freq*time) ; A real part
.measure Aim 2*(V(out)-Aavg)*sin(360*Freq*time) ; A imag part
.measure Bre 2*(V(in)-Bavg)*cos(360*Freq*time) ; B real part
.measure Bim 2*(V(in)-Bavg)*sin(360*Freq*time) ; B imag part
.measure Amag hypot(Areal,Aim) ; A magnitude
.measure Bmag hypot(Breal,Bim) ; B magnitude
.measure Aphi param atan2(Areal,Aim) ; A phase
.measure Bphi param atan2(Breal,Bim) ; B phase
.measure GainMag param 20*log10(Amag/Bmag) ; Gain magnitude
.measure GainPhi param Bphi - Aphi ; Gain phase

```

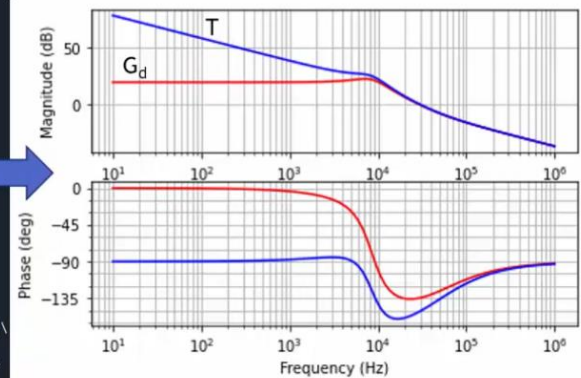




```

1 # -*- coding: utf-8 -*-
2 from math import pi, log10, sqrt
3 from control import tf, bode_plot, margin
4 L= 10e-6; rL=0.05; C=33e-6; rC=0.1; R=1; Vi= 10; D=0.53;
5 # PWM modulator gain
6 kd=1
7 # PI compensator
8 R1=1.81e3; R2=1.81e3; C2=10e-9
9 wz=1/(R2*C2)
10 # Transfer functions
11 s = tf('s')
12 # Control-to-output
13 Gd = Vi*(1+rC/R)*s**2 + (L/R+rC*C+rL*C+rC*C/R)*s + 1 + rL/R
14 Cp=(1+s/wz)/(s/wz) # Compensator
15 H=1 # Sensor
16 # Open-loop audiosusceptibility
17 Gi= D*(1 + rC*C*s)/ \
18     (L*C*(1+rC/R)*s**2 + (L/R+rC*C+rL*C+rC*C/R)*s + 1 + rL/R)
19 # Loop gain
20 T=Cp*kd*Gd*H
21 # Plot Plant's Bode
22 # Note that once Hz is true, omega limits are in Hz
23 mag, phase, omega = bode_plot(Gd, dB=True, Hz=True, omega_limits=(10,1000e3),\
24                               omega_num=100, color="red")
25 mag, phase, omega = bode_plot(T, dB=True, Hz=True, omega_limits=(10,1000e3),\
26                               omega_num=100, color="blue")
27 # Print phase margin and crossover frequency
28 gm, pm, wcp, wcp = margin(T)
29 print ("Phase margin= ", pm)
30 print ("Crossover frequency (0 dB)= ", wcp/(2*pi));

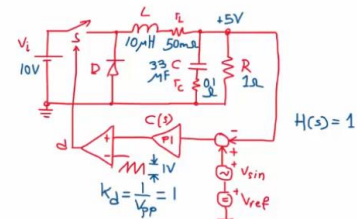
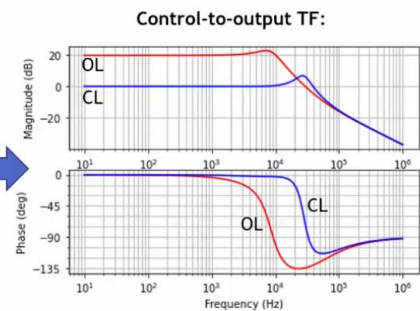
```



```

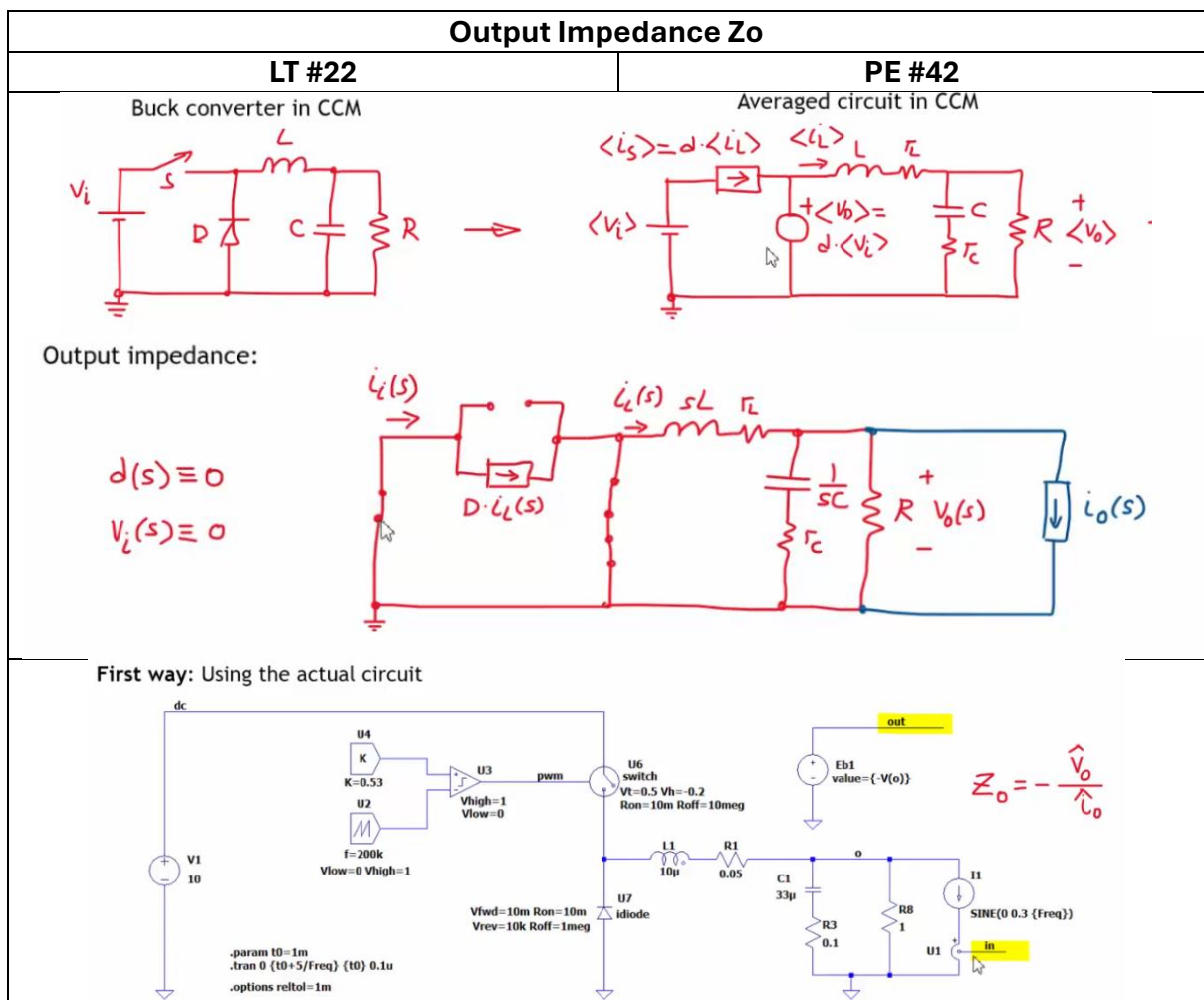
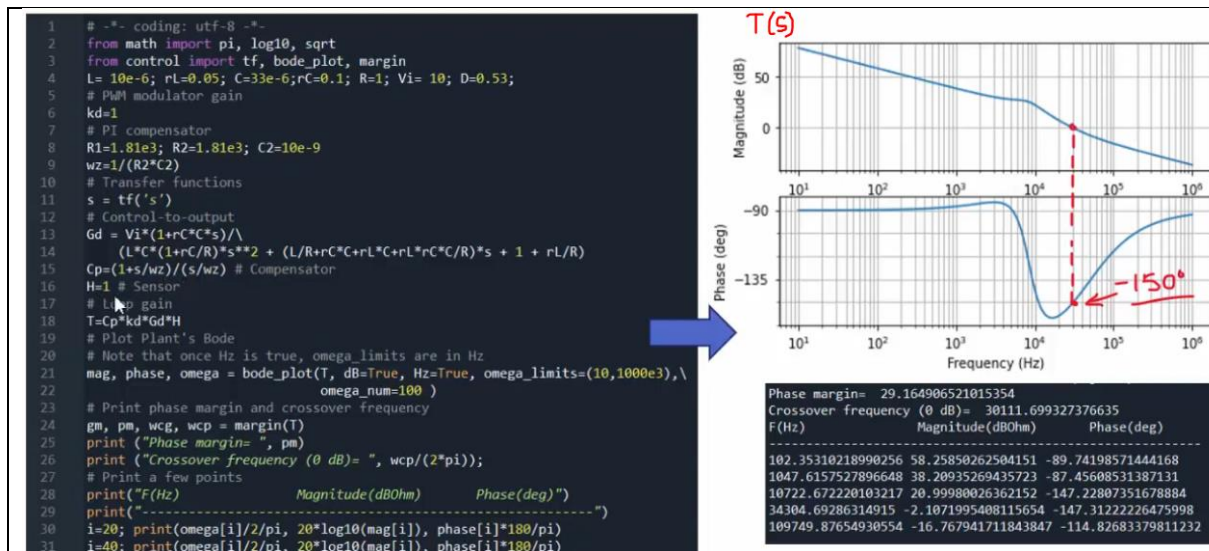
1 # -*- coding: utf-8 -*-
2 from math import pi, log10, sqrt
3 from control import tf, bode_plot
4 L= 10e-6; rL=0.05; C=33e-6; rC=0.1; R=1; Vi= 10; D=0.53;
5 # PWM modulator gain
6 kd=1
7 # PI compensator
8 R1=1.81e3; R2=1.81e3; C2=10e-9
9 wz=1/(R2*C2)
10 # Transfer functions
11 s = tf('s')
12 # Control-to-output
13 Gd = Vi*(1+rC/R)*s**2 + (L/R+rC*C+rL*C+rC*C/R)*s + 1 + rL/R
14 Cp=(1+s/wz)/(s/wz) # Compensator
15 H=1 # Sensor
16 # Loop gain
17 T=Cp*kd*Gd*H
18 # Closed-loop response
19 Gcl= Cp*kd*Gd/(1+T)
20 # Plot Plant's Bode
21 # Note that once Hz is true, omega limits are in Hz
22 mag, phase, omega = bode_plot(Gd, dB=True, Hz=True, omega_limits=(10,1000e3),\
23                               omega_num=100, color="red")
24 mag, phase, omega = bode_plot(Gcl, dB=True, Hz=True, omega_limits=(10,1000e3),\
25                               omega_num=100, color="blue")

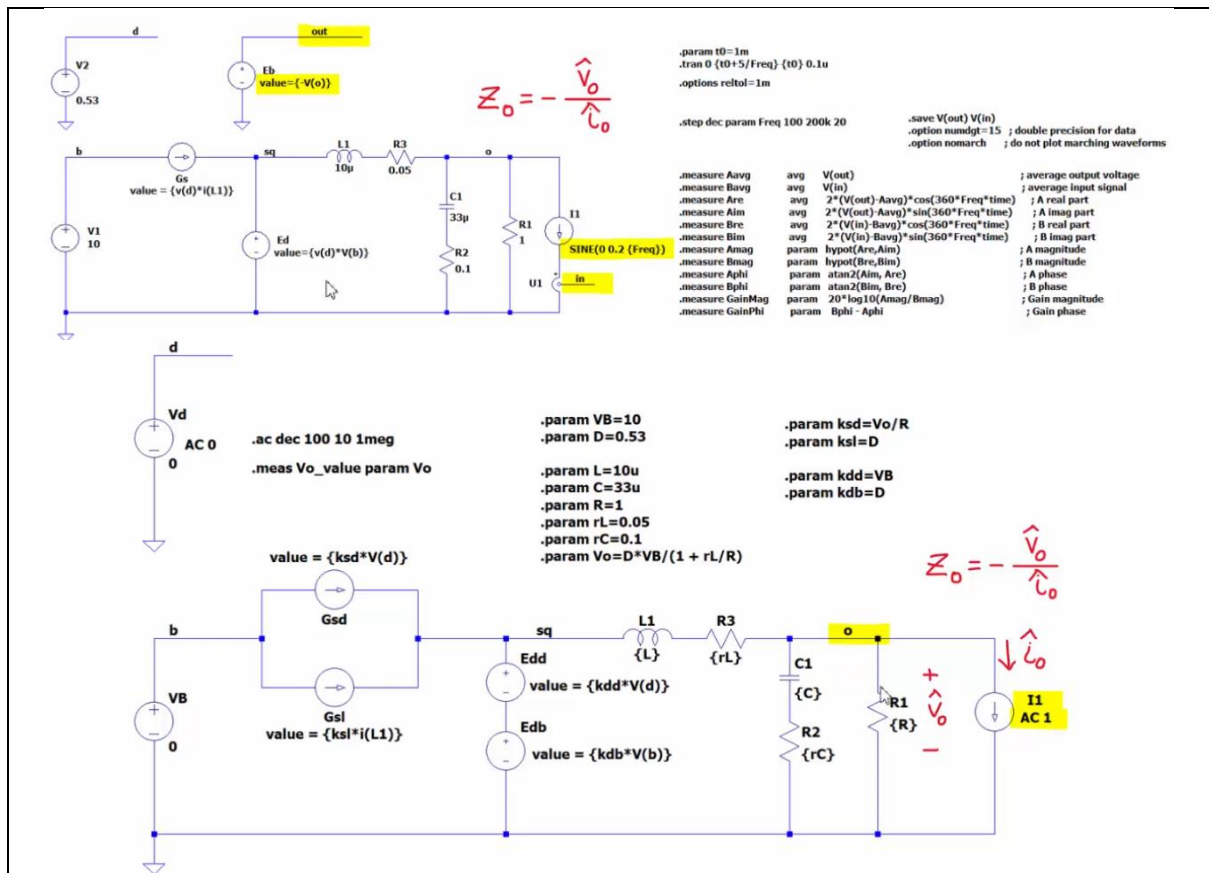
```



$$G_{CL} = \frac{V_o(s)}{V_{ref}(s)} = \frac{C(s) \cdot K_d \cdot G_d(s)}{1 + C(s) \cdot K_d \cdot G_d(s) \cdot H(s)}$$

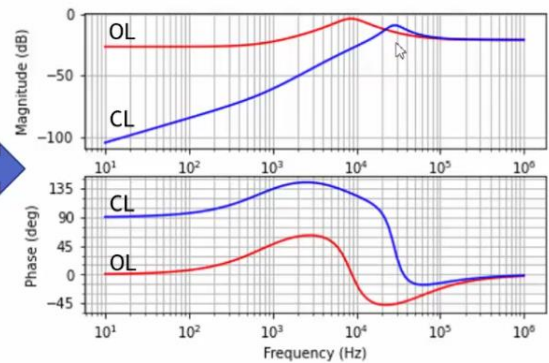
$T(s)$





Output impedance: $Z_o(s)$

```
1 # -*- coding: utf-8 -*-
2 from math import pi, log10, sqrt
3 from control import tf, bode_plot
4 L= 10e-6; rL=0.05; C=33e-6; rC=0.1; R=1; Vi= 10; D=0.53;
5 # PWM modulator gain
6 kd=1
7 # PI compensator
8 R1=1.81e3; R2=1.81e3; C2=10e-9
9 wz=1/(R2*C2)
10 # Transfer functions
11 s = tf('s')
12 # Control-to-output
13 Gd = Vi*(1+rC*C*s)/\
14     (L*C*(1+rC/R)*s**2 + (L/R+rC*C+rL*L*rC*C/R)*s + 1 + rL/R)
15 Cp=(1+s/wz)/(s/wz) # Compensator
16 H=1 # Sensor
17 # Open-loop output impedance
18 Zo= (rL+L*s)*(1+rC*C*s)/\
19     (L*C*(1+rC/R)*s**2 + (L/R+rC*C+rL*L*rC*C/R)*s + 1 + rL/R)
20 # Loop gain
21 T=Cp*kd*Gd*H
22 # Closed-loop output impedance
23 Zo_cl= Zo/(1+T)
24 # Plot Plant's Bode
25 # Note that once Hz is true, omega limits are in Hz
26 mag, phase, omega = bode_plot(Zo, dB=True, Hz=True, omega_limits=(10,1000e3),\
27     omega_num=100, color='red')
28 mag, phase, omega = bode_plot(Zo_cl, dB=True, Hz=True, omega_limits=(10,1000e3),\
29     omega_num=100, color='blue')
30
```



Audio susceptibility

